



Department of Physical Sciences
Faculty of Applied Sciences
Rajarata University of Sri Lanka, Mihintale
B.Sc. in Applied Sciences

STA 1303 – Probability Theory
Tutorial No: 2

1. A box contains 5 red balls, 4 white balls, and 3 blue balls. A ball is selected at random from the box, its color is noted, and then the ball is replaced. Find the probability that out of 6 balls selected in this manner, 3 are red, 2 are white, and 1 is blue.
2. When commercial aircraft are inspected, wing cracks are reported as nonexistent, detectable, or critical. The history of a particular fleet indicates that 70% of the planes inspected have no wing cracks, 25% have detectable wing cracks, and 5% have critical wing cracks. Five planes are randomly selected. Find the probability that
 - a. one has a critical crack, two have detectable cracks, and two have no cracks.
 - b. at least one plane has critical crack

3. Let Y be a random variable with probability density function given by

$$f(Y) = \begin{cases} 2(1 - Y), & 0 \leq Y \leq 1, \\ 0 & \text{elsewhere} \end{cases}$$

- a. Find the density function of $U_1 = 2Y - 1$.
 - b. Find the density function of $U_2 = 1 - 2Y$.
 - c. Find the density function of $U_3 = Y^2$.
 - d. Find $E(U_1)$, $E(U_2)$, and $E(U_3)$ by using the derived density functions for these random variables.
4. Let Y be a random variable with a density function given by

$$f(Y) = \begin{cases} \left(\frac{3}{2}\right)Y^2, & -1 \leq Y \leq 1, \\ 0 & \text{elsewhere} \end{cases}$$

- a. Find the density function of $U_1 = 3Y$.
 - b. Find the density function of $U_2 = 3 - Y$.
 - c. Find the density function of $U_3 = Y^2$.

5. The amount of flour used per day by a bakery is a random variable Y that has an exponential distribution with mean equal to 4 tons. The cost of the flour is proportional to $U = 3Y$
 - a. Find the probability density function for U .
 - b. Use the answer in part (a) to find $E(U)$.
6. Let Y_1 and Y_2 be independent random variables with moment-generating functions $m_{Y_1}(t)$ and $m_{Y_2}(t)$, respectively. If a_1 and a_2 are constants, and $U = a_1Y_1 + a_2Y_2$ show that the moment-generating function for U is $m_U(t) = m_{Y_1}(a_1t) \times m_{Y_2}(a_2t)$.
7. Electronic components of a certain type have a length of life Y , with probability density given by

$$f(Y) = \begin{cases} 2\left(\frac{1}{100}\right)e^{\frac{-y}{100}}, & Y > 0, \\ 0 & \text{elsewhere} \end{cases}$$

(Length of life is measured in hours.) Suppose that two such components operate independently and in series in a certain system (hence, the system fails when either component fails). Find the density function for X , the length of life of the system

(Please answer question number 1, 2, 3, 4, and 7 and submit the answer script to the tutorial room on or before 18th of October 2023)